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TRANSCATHETER HEART VALVE WITH DEFORMABLE INDUCTOR FOR DUAL WIRELESS PRESSURE MONITORING

Abstract

A prosthetic valve includes a flexible frame disposed along and deformable about a frame axis. The frame includes a first end, a second end oppositely disposed from the first end, and a network of interconnected struts defining a plurality of cells. A first circuit is mounted on the frame. The first circuit includes a first inductor coil attached to and tracing a first subset of the struts such that the first inductor coil outlines a first subset of the plurality of cells, and a first sensor in electrical communication with the first inductor coil. The first sensor is configured to sense a physical parameter and generate a signal representing the physical parameter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of International Application No. PCT/US2023/084312, filed Dec. 15, 2023, which claims the benefit of U.S. Provisional Application No. 63/387,915, filed Dec. 16, 2022, the disclosures of which are hereby incorporated by reference in their entireties.

BACKGROUND

[0002] The present disclosure relates to medical implant devices, and in particular, to implantable prosthetic valves.

[0003] Patients who receive heart valve implants can suffer from post-operation complications. Risk of complications is especially high within thirty or sixty days of an implant operation. However, during such periods of time, the patient may no longer be in a hospital or extended care facility/system, and therefore complications that arise may require reentry into the care system, potentially adding significant cost to the overall patient treatment. Furthermore, increased health risks may result from the patient delaying return to the hospital due to failure to recognize the complications until they manifest through perceivable symptoms that the patient interprets as requiring hospital care. Accordingly, systems, devices and methods for post-operatively monitoring prosthetic heart valve implant recipients, including in an environment outside of the hospital or care facility, are desirable for improving patient outcomes

SUMMARY

[0004] In one example, a prosthetic valve includes a flexible frame disposed along and deformable about a frame axis. The frame includes a first end, a second end oppositely disposed from the first end, and a network of interconnected struts defining a plurality of cells. A first circuit is mounted on the frame. The first circuit includes a first inductor coil attached to and tracing a first subset of the struts such that the first inductor coil outlines a first subset of the plurality of cells, and a first sensor in electrical communication with the first inductor coil. The first sensor is configured to sense a physical parameter and generate a signal representing the physical parameter.

[0005] In another example, a prosthetic valve assembly includes a prosthetic valve with a flexible frame disposed along a frame axis and deformable about the frame axis between a crimped state and an expanded state. The frame includes a first end, a second end oppositely disposed from the first end, and a network of interconnected struts defining a plurality of cells. A first circuit is mounted on the frame. The first circuit includes a first inductor coil attached to and tracing a first subset of the struts such that the first inductor coil outlines a first subset of the plurality of cells, and a first sensor in electrical communication with the first inductor coil. The first sensor is configured to sense a physical parameter and generate a signal representing the physical parameter. The prosthetic valve assembly further includes a transmitter in communication with the first sensor.

[0006] In another example, a prosthetic valve includes a flexible frame disposed along a frame axis and deformable about the frame axis between a crimped state and an expanded state. The frame includes a first end, a second end oppositely disposed from the first end, and a network of interconnected struts defining a plurality of cells. A multilayer sensing assembly is mounted on the frame. The multilayer sensing assembly includes a first inductor coil pair comprising first upper and lower inductor coil portions, the first inductor coil pair disposed on a flexible substrate, a detuning mitigation layer disposed between the frame and the flexible substrate, and a first sensor

in electrical communication with the first inductor coil pair. The first sensor is configured to sense a physical parameter and generate a signal representing the physical parameter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a schematic illustration of a human patient with a heart.

[0008] FIG. 2 is a partial cross-sectional schematic of the heart.

[0009] FIG. 3 is a block diagram representing a monitoring system for monitoring one or
[0010] more physiological parameters associated with a patient.

[0011] FIG. 4 is a perspective view of a first example of a prosthetic heart valve shown in an expanded state.

[0012] FIG. 5 is a front view of a sensing circuit of the prosthetic heart valve of FIG. 4, shown in isolation from the frame.

[0013] FIG. 6 is a schematic cross-sectional exploded view of a strut and an inductor coil of the prosthetic heart valve of FIG. 4 taken transverse to axis A of FIG. 4.

[0014] FIG. 7 is a perspective view of a second example of a prosthetic heart valve shown in the expanded state.

[0015] FIG. 8 is a schematic front view of a third example of a prosthetic heart valve shown in a crimped state.

[0016] FIG. 9 is a schematic plan view comparing radial dimensions of a frame of the third example of the prosthetic heart valve in the expanded state and the crimped state.

[0017] FIG. 10 is a schematic view of a fourth example of a prosthetic heart valve shown in the expanded state with a fabric cover over the inductor coil.

[0018] FIG. 11 is a schematic cross-sectional view showing a portion a multilayer sensing assembly disposed on a flexible frame.

DETAILED DESCRIPTION

[0019] FIG. 1 is an anterior view of human patient 2 with heart 4. The body of patient 2 can generally be bisected by any of three planes: a coronal (i.e., x-y) plane, a sagittal (i.e., y-z) plane, and a transverse (i.e., x-z) plane.

[0020] FIG. 2 is a partial cross-sectional schematic of heart 4. Heart 4 includes four chambers, including left atrium 6, left ventricle 8, right ventricle 10, and right atrium 12. The four chambers are shown in cross-section in FIG. 2. Heart 4 further includes four valves for aiding the circulation of blood therein, including tricuspid valve 14, pulmonary valve 16, mitral valve 18, and aortic valve 20. FIG. 2 further shows pulmonary artery 21 and aorta 22.

[0021] Tricuspid valve 14 separates right atrium 12 from right ventricle 10 and can include three cusps or leaflets. Tricuspid valve 14 can close during ventricular contraction (i.e., systole) and open during ventricular expansion (i.e., diastole). Pulmonary valve 16 separates right ventricle 10 from pulmonary artery 21 and may be configured to open during systole so that blood may be pumped towards the lungs, and close during diastole to prevent blood from leaking back into heart 4 from pulmonary artery 21. Similar to tricuspid valve 14, pulmonary valve 16 can have three cusps/leaflets, each one resembling a crescent. Mitral valve 18 separates left atrium 6 from left ventricle 8 and can have two cusps or leaflets. Mitral valve 18 is configured to open during diastole so that blood in left atrium 6 can flow into left ventricle 8, and close during systole to prevent blood from leaking back into left atrium 6. Aortic valve 20 separates left ventricle 8 from aorta 22. Aortic valve 20 is configured to open during systole to allow blood leaving left ventricle 8 to enter aorta 22, and close during diastole to prevent blood from leaking back into left ventricle 8.

[0022] A heart valve can include a relatively dense fibrous ring, referred to herein as the annulus, as well as a plurality of leaflets or cusps attached to the annulus. Some valves can further include a

collection of chordae tendineae and papillary muscles securing the leaflets. Generally, the size of the leaflets or cusps may be such that when the heart contracts, the resulting increased blood pressure produced within the corresponding heart chamber forces the leaflets open at least partially to allow flow from the heart chamber. As the pressure in the heart chamber subsides, the pressure in the subsequent chamber or blood vessel may become dominant and press back against the leaflets. As a result, the leaflets/cusps come in apposition to each other, thereby closing the flow passage. [0023] Heart valve disease represents a condition in which one or more of the valves of heart **4** fails to function properly. Diseased heart valves may be categorized as stenotic, wherein the valve does not open sufficiently to allow adequate forward flow of blood through the valve, and/or incompetent, wherein the valve does not close completely, causing excessive backward flow of blood through the valve when the valve is closed. In certain conditions, valve disease can be severely debilitating and even fatal if left untreated.

[0024] To treat disease of, for example, mitral valve **18**, a prosthetic heart valve can be implanted in and sutured to the annulus of mitral valve **18**. Such a prosthetic heart valve can be positioned with its openings oriented in the direction of blood flow from left atrium **6** to left ventricle **8**. The prosthetic heart valve can be configured to operate as aortic valve **20** such that it can allow unidirectional blood flow left ventricle **8** from left atrium **6** while preventing flow in the reverse direction.

[0025] In a typical cardiac implant procedure, the heart can be incised, and in a valve replacement operation, the defective valve can be removed leaving the desired placement site that can include the valve annulus. Sutures can be passed through fibrous tissue of the annulus or desired placement site to form an array of sutures. Free ends of the sutures may be individually threaded through a suture-permeable sealing edge of the prosthetic heart valve. Artificial heart valves can be used to replace faulty or deteriorating natural heart valves in patients with heart valve disorders including aortic stenosis, mitral regurgitation, etc. The valve replacement process generally involves surgical or transcatheter procedures (e.g., balloon valvotomy) to replace the existing valves with the new artificial valves. Since the artificial valves are a foreign body, many different challenges and issues can be involved with such a procedure. For example, paravalvular leakage (PVL) and/or leaflet thickening can occur in patients who undergo heart valve replacement. Similarly, rejection of an artificial surgical heart valve due to thrombus can occur, requiring the patient to use anti-coagulants for proper valve operation.

[0026] Some methods for monitoring valve performance after implantation involve using complex bio-imaging techniques, such as echocardiography. Such methods can generally only be performed in specialized medical facilities and can cost significant time and money. Hence, such methods may generally only be used once symptoms of valve malfunction are detected. Some artificial valves may not provide an ability to detect changes in operation to detect problems early on. Moreover, many patients who suffer from valvular disease and require an artificial valve may also suffer from other cardiovascular disorders, including heart failure. Some artificial heart valve systems may not allow for gathering data about the valve and/or the patient's condition postoperatively in an outpatient setting (e.g., a cardiologist visit in a ward) using existing patient monitoring systems. Such systems may not provide for routine collection of data at sufficient resolution to enable development of new digital solutions for better management of the patients as their numbers and diversity increase over time.

[0027] Accordingly, a prosthetic heart valve can be part of a larger system for post-operatively monitoring a patient, as will be discussed in reference to FIG. **3**.

[0028] FIG. **3** is a block diagram representing monitoring system **23** for monitoring one or more physiological parameters associated with a patient (e.g., patient **2** shown in FIG. **1**). System **23** includes prosthetic heart valve **24**, which includes sensing devices **26**, control circuitry **28**, transmitter **30**, and power source **32**. System **23** further includes external device **34**, which includes antenna **36**, control circuitry **38**, and transceiver **40**. System **23** also includes cloud **42** and remote

monitor **44**.

[0029] Prosthetic heart valve **24** can include one or more sensing devices **26**, control circuitry **28**, transmitter **30**, and power source **32**. Sensing devices **28** can include one or more of following types of sensors/transducers: MEMS sensors, optical sensors, piezoelectric sensors, electromagnetic sensors, strain sensors/gauges, accelerometers, gyroscopes, and/or other types of sensors, which can be positioned in the patient to sense one or more parameters relevant to the health of the patient. Control circuitry **28** can be wired or wirelessly connected to sensing devices **26** and can include one or more of application-specific integrated circuit (ASIC), microcontrollers, chips, tuning capacitors, etc. Control circuitry **28** can receive signals from external device **34** (e.g., requests for stored or immediately acquired data), request data from sensors **26**, and coordinate data transmission. Transmitter **30** can be, for example, an antenna for radiating an electronic signal transmitted by control circuitry **28**. Power source **32** can be a suitable source of power able to minimize interference with the heart or other anatomy of the patient. In one example, power source **32** can be a passive means for wirelessly receiving external power (e.g., short-range or near-field wireless power transmission). In another example, power source **32** can be a battery, or a means for locally harvesting energy from within the patient.

[0030] External device **34**, located at least partially outside of the patient, can be in wireless communication with prosthetic heart valve **24**. External device **34** includes antenna **36**, control circuitry **38**, and transceiver **40**. Antenna **36** can receive wireless signal transmissions from prosthetic heart valve **24**. In one example, antenna **36** can be externally mounted to external device **34**. Control circuitry **38** can be a processor or other suitable means for processing signals received from prosthetic heart valve **24**. Transceiver **40** can be configured to receive and amplify signals from prosthetic heart valve **24**, as well as to transmit signals to cloud **42** and remote monitor **44**. Such signals can include, for example, pressure data acquired from sensors **26**. Transceiver **40** can, accordingly, include one or more of digital-to-analog converter (DAC) circuitry, power amplifiers, low-pass filters, antenna switch modules, antennas, etc. for treatment and/or processing of transmitted and received signals.

[0031] External device **34** can serve as an intermediate communication device between prosthetic heart valve **24** and remote monitor **44**. External device **34** can be a dedicated external unit designed to communicate with prosthetic heart valve **24**. For example, external device **34** can be a wearable communication device, or other device that can be readily disposed in proximity to the patient and/or prosthetic heart valve **24**. External device **34** can be configured to interrogate prosthetic heart valve continuously, periodically, or sporadically **24** in order to extract or request sensor-based information therefrom. In some examples, external device **34** can include a user interface upon which a user (e.g., the patient) can view sensor data, request sensor data, or otherwise interact with external device **34** and/or prosthetic heart valve **24**.

[0032] Cloud **42** can be a secure network in communication with external device **34** via ethernet, Wi-Fi, or other network protocol. Cloud **42** can also be configured to implement data storage. In another example, cloud **42** can instead be a secure physical network. Remote monitor **44** can be in communication with external device **34** via cloud **42**. Remote monitor **44** can be any type of computing device or collection of computing devices configured to receive, process and/or present monitor data received via cloud **42** from external device **34** or prosthetic heart valve **24**. For example, remote monitor **44** can advantageously be operated and/or controlled by a healthcare entity, such as a hospital, doctor, or other care entity associated with the patient. Although certain examples disclosed herein describe communication with remote monitor **44** from prosthetic heart valve **24** indirectly through external device **34**, prosthetic heart valve **24** can instead include a transmitter (e.g., transmitter **30**) capable of communicating, via cloud **42**, with remote monitor **44** without the necessity of relaying information through device **34**.

[0033] FIG. **4** is a perspective view of prosthetic heart valve **124** shown in an expanded state. FIG. **5** is a front view of a sensing circuit of prosthetic heart valve **124**, shown in isolation from the

frame. FIGS. 4 and 5 are discussed together.

[0034] As shown in FIG. 4, structural components of prosthetic heart valve **124** include deformable frame **146** and post assemblies **148** extending axially away from frame **146** relative to valve axis A. Axis A can generally be aligned with the direction of blood flow through prosthetic heart valve **124** when implanted in heart **4**. Frame **146** can be formed from a biocompatible metallic material. As shown in FIG. 4, one post assembly **148** extends from each of top/upper end **150** and bottom/lower end **152** of prosthetic heart valve **124** based on the orientation of FIG. 4. Each post assembly **148** can include post **154** and islet **156** upon which sensor **126** can be mounted. As shown in FIG. 4, islet **156** can have a generally square shape corresponding to the shape of sensor **126**. Frame **146** comprises a network of struts **158** defining open cells **160** therebetween. Each cell **160** can include oppositely axially disposed pointed tips/ends **162**.

[0035] Electrical components of prosthetic heart valve **124** include one or more sensing circuits **164** for monitoring physiological parameters of patient **2**. Each sensing circuit **164** includes deformable inductor coil **166** and sensor **126** electrically connected (e.g., via leads/wires) to inductor coil **166**. Sensing circuit **164** can be an inductor-resistor-capacitor (LCR) circuit **168**, with inductor coil **166** forming the inductor (L) and resistor (R) elements of circuit **168**, and sensor **126**, connected in parallel, forming the capacitor (C) element. Each LCR circuit **168** of prosthetic heart valve **126** has a distinct self-resonant frequency. The self-resonant frequency for each circuit can be represented as $f=1/2\pi\sqrt{LC(p)}$, where L is the inductance of inductor coil **166** and C(p) is the capacitance of sensor **126** at a given pressure. In general, the self-resonant frequency for each LCR circuit **168** can range from 5 MHz to 50 MHz, and more specifically, from 10 MHz to 20 MHz.

[0036] Inductor coil **166** can include one or more individual wires formed from a conductive, but biocompatible, metallic material, such as gold. Other examples can include copper or titanium. Inductor coil **166** can further be coated with an insulating coating (shown and labeled in FIG. 6). Sensors **126** can be capacitive pressure sensors in one example, each including a diaphragm and pressure cavity to form a variable capacitor to detect strain due to pressure applied to the diaphragm. In general, the capacitance of sensors **126** decreases as pressure deforms the diaphragms. To manage detuning of sensing circuit **164**, a detuning mitigation layer, discussed in greater detail below with respect to FIGS. 6 and 11, can be positioned between inductor coil **166** and struts **158** of frame **146**.

[0037] Inductor coil **166** can be removably attached to frame **146** by sutures **170**, shown schematically in FIG. 5. Sutures **170** can be formed from a biocompatible polymer in one example. More specifically, inductor coil **166** can be attached to frame **146** in such manner as to trace a subset of struts **158** and outline a subset of cells **160**. In this regard, inductor coil **166** can have nearly identical geometric attributes to struts **158** and cells **160**, for example, having pointed tips **172** corresponding to pointed tips **162** of the underlying cells **160** of frame **146**. In the example of FIGS. 4 and 5, inductor coil **166** can be disposed to trace/frame/outline a two by three subset (i.e., two cells high in the axial direction and three cells long in a radial dimension) of cells **160** of frame **146**. This can include uppermost or lowermost cells, along with interior cells **160**. Other arrangements are contemplated herein. Sutures **170** can be disposed at various points along inductor coil **166** to ensure that inductor coil **166** is secured to and maintains the shape of the supporting subset of struts **158**. Suture points can include pointed tips **162** of cells **160** of frame **146** and pointed tips **172** of inductor coil **166**, respectively. Additional and/or alternative suture points are contemplated herein.

[0038] FIG. 6 is a schematic cross-sectional exploded view of strut **158** and inductor coil **166** of the prosthetic heart valve of FIG. 4 taken transverse to axis A of FIG. 4. FIG. 6 shows strut **158**, inductor coil **166**, ferrite layer **174**, insulating layer **176**, and adhesive backing layer **178**.

[0039] As shown in FIG. 6, strut **158** is the innermost layer, and inductor coil **166** is the outermost layer. Ferrite layer **174** is disposed between strut **158** and inductor coil **166**. Ferrite layer **174** can be formed as a strip of a soft, ferromagnetic material in one example, and can mitigate detuning

caused by the proximity of inductor coil **166** to metallic strut **158**. Ferrite layer **174** can additionally improve (e.g., increase) the sensing range of sensing circuit **164** by shielding inductor coil **166** from magnetic field interference, induced eddy currents, etc. caused generally by electronic components of prosthetic heart valve **124**. Ferrite layer **174** is ideally coextensive with inductor coil **166** such that no area of inductor coil **166** is exposed to the underlying metallic strut(s) **158**. Additionally, the ferrite layer **174** should be sufficiently flexible to transition between crimped and expanded states of prosthetic heart valve **124** and maintain its position between struts **158** and inductor coil **166**. Insulating layer **176** surrounds inductor coil **166** and ferrite layer **174**. Ferrite layer **174** and inductor coil **166** can be in direct physical contact, (e.g., as an integrated layer) and insulating layer **176** can encircle/surround inductor coil **166** and ferrite layer **174**. In one example, insulating layer **176** can be a biocompatible elastomer (e.g., silicone) or polymer (e.g., parylene or polyimide). In some examples, adhesive backing layer **178** can be included between insulating layer **176** and strut **158**. Such an adhesive layer can be both biocompatible and non-conductive. [0040] FIG. 7 is a perspective view of prosthetic heart valve **224** shown in an expanded state. Prosthetic heart valve **224** is substantially similar to prosthetic heart valve **124** shown in FIG. 4, having deformable frame **246** and post assemblies **248** extending axially away from frame **246** at each of upper end **250** and lower end **252**. Post assembly **248** includes post **254** and islet **256** for mounting sensor **226** thereupon. Struts **258** of frame **246** define cells **260**. Cells **260** can include oppositely disposed pointed tips **262**. Prosthetic heart valve **224** further includes two sensing circuits **264**, configured as LCR circuits **268**, each including sensor **226** and inductor coil **266** tracing a subset of struts **258** and outlining a subset of cells **260**. Inductor coil **266** can be formed from one or more conductive (e.g., gold) wires and include an underlying ferrite layer (not shown in FIG. 7). Sutures (not shown in FIG. 7) can secure each inductor coil **266** to frame **246** in the manner discussed above with respect to FIGS. 4 and 5.

[0041] Unlike the foregoing example, prosthetic heart valve **224** includes biocompatible fabric **280**, configured as a skirt and partially covering frame **246**. Fabric **280** can be formed from a polymer material. In an alternative example, fabric **280** can fully cover frame **246** such that no struts **258** are exposed on the outer side of frame **246**. Prosthetic heart valve **224** also includes pericardium tissue **282** which can be formed from a synthetic material or derived from a mammalian (e.g., bovine) tissue source.

[0042] FIG. 8 is a schematic front view of prosthetic heart valve **324** shown in a crimped state. Prosthetic heart valve **324** is substantially similar to prosthetic heart valve **124** shown in FIG. 4 and prosthetic heart valve **224** shown in FIG. 7, with deformable frame **346** having interconnected struts **358** defining cells **360**. Frame **346** can be formed from a biocompatible metallic material. Post assembly **348** (only one is shown in FIG. 8) extends axially away from upper end **350**. Prosthetic heart valve **324** can further include at least one sensing circuit **364**, with a flexible inductor coil **366** in electrical communication with sensor **326** via wire **392**. In the crimped state, the axial dimension of frame **346** (i.e., along axis A) is greater than when in the expanded state, such that upper end **350** is further from lower end **352** in the crimped state. Further, in the crimped state, inductor coil **366** sutured to frame **346** will deform similarly to frame **346**, maintaining the geometry of the underlying struts **358** and framing the deformed cells **360**, and increasing/elongating in the axial direction. This can occur, for example, because of the relative overall flexibility of the inductor coil material and/or surrounding layers (e.g., insulating layer and detuning mitigation means), the thicknesses of the layers, and the robustness of attachment means (e.g., sutures) for securing the inductor coil to the underlying struts.

[0043] FIG. 9 is a schematic plan view comparing radial dimensions of frame **346** of prosthetic heart valve **324** in the expanded state and the crimped state. Frame **346** is the frame of prosthetic heart valve **324** shown in FIG. 8. Frame **346** in the expanded state is represented by the solid line, while frame **346** in the crimped state is represented by the dashed line. As shown in FIG. 9, in the expanded state, frame **346** has a radius R_1 , while in the crimped state, frame has a radius R_2 . R_1 is

greater than R2. As such, in the expanded state, the axial dimension of frame **346** is less than the axial dimension of frame **346** in the crimped state. In the expanded state, the radial dimension (i.e., radius R1) of frame **346** is greater than the radial dimension (i.e., radius R2) of frame **346** in the crimped state.

[0044] FIG. **10** is a schematic perspective view of prosthetic heart valve **424** shown in an expanded state with fabric cover **484** over inductor coil **466**. For simplicity, prosthetic heart valve **424** is shown without post assemblies and sensors. Prosthetic heart valve **424** is substantially similar to prosthetic heart valve **124** shown in FIG. **4**, prosthetic heart valve **224** shown in FIG. **7**, and prosthetic heart valve **324** shown in FIG. **8**. Prosthetic heart valve **424** has a biocompatible metallic frame **446** with interconnecting struts defining cells (not shown in FIG. **10**), an upper end **450**, and lower end **452**. Prosthetic heart valve **424** further includes sensing circuits **464** (only one is shown) comprising inductor coil **466** and a sensor (not shown). Inductor coil **466** can be formed of one or more wires of a conductive material (e.g., gold). Sutures **470** can secure inductor coil **466** to struts of frame **446**.

[0045] Prosthetic heart valves **424** differs from the previous examples in that it includes cover **484** disposed over inductor coil **466** and sutures **470**. The concealed components are, accordingly, represented by dashed lines. Cover **484** can be a biocompatible fabric substantially similar to fabric **280** of FIG. **7**. Cover **484** can be stitched and/or woven to frame **446** to secure it in place in one example, while in another example, cover **484** can be arranged as a sleeve circumscribing frame **446**. Cover **484** can protect the concealed components from native tissue ingrowth after implantation of prosthetic heart valve **424**. Cover **484** can further protect the surrounding tissues from catching on/coming into direct contact with the concealed components.

[0046] FIG. **11** is a schematic cross-sectional view of a portion of multilayer sensing assembly **586** disposed on flexible frame **546**. Multilayer sensing assembly **586** incorporates multiple inductor coils **566** stacked/layered in the radial direction, relative to the axis of the encompassing prosthetic heart valve and can be used with any of the prosthetic heart valves disclosed herein.

[0047] As shown in FIG. **11**, multilayer sensing assembly **586** is disposed upon metallic strut **558**, which can be formed from a biocompatible metallic material. Multilayer sensing assembly **586** includes first inductor coil pair **566A** and second inductor coil pair **566B**, each including upper and lower coil portions printed on flexible substrate **588**. Flexible substrate **588** can be formed from a polymer material, such as polyimide, and can have a thickness ranging from 2 millimeters to 3 millimeters in one example. Each inductor coil pair **566A** and **566B** can be printed onto flexible substrate **588**, and can be formed from gold in one example, or from copper or titanium in an alternative example. Each inductor coil pair **566A** and **566B** can be electrically connected to a respective sensor **526**, represented schematically in FIG. **11**, to form sensing circuits **564**. Sensors **526** can be substantially similar to sensor **126** shown in FIGS. **4** and **5**, sensor **226** shown in FIG. **7**, and/or sensor **326** shown in FIG. **8**. Each upper coil portion of inductor coil pairs **566A** and **566B** can further be electrically connected (e.g., through conductive vias) to a respective lower coil portion. Inductor coil pairs **566A** and **566B** can be arranged to follow struts **558** and cells (not shown), having a pattern similar to inductor coil **166** shown in FIGS. **4-6**, inductor coil **266** shown in FIG. **7**, inductor coil **366** shown in FIG. **8**, and inductor coil **466** shown in FIG. **10**. Inductor coil pairs **566A** and **566B** can alternatively be arranged as rounded or square spirals in another example.

[0048] Multilayer sensing assembly **586** further includes flexible ferrite layer **574** disposed between strut **558** and inductor coil pairs **566A** and **566B**. Ferrite layer **574** can be substantially similar to ferrite layer **174** shown in FIG. **6** in that it prevents magnetic field interference and mitigates detuning of sensing circuits **564**. Exemplary materials for ferrite layer **574** can include manganese-zinc (MnZn) ferrites or nickel-zinc (NiZn) ferrites. In the example shown in FIG. **11**, ferrite layer **574** can be printed onto flexible substrate **588**. Ferrite layer **574** can have a thickness ranging from 0.25 millimeters to 0.35 millimeters in one example. Soft magnetic layers **590A** and **590B** are further disposed between upper and lower coil portions of respective inductor coil pair

566A and **566B**, and can be formed from frequency-dependent soft magnetic materials. More specifically, soft magnetic layer **590A** can be responsive and tuned to the frequency at which inductor coil pair **566A** is operating. Similarly, soft magnetic layer **590B** can be responsive and tuned to the frequency at which inductor coil pair **566B** is operating. Soft magnetic layers **590A** and **590B** thus isolate the two, different frequencies of their respective inductor coil pairs, minimizing interference and crosstalk between inductor coil pairs **566A** and **566B**. Ferrite layer **574**, flexible substrate **588**, and the various layers disposed therein can be encapsulated by insulating layer **576**, which, like insulating layer **176** shown in FIG. 6, can be a biocompatible silicone, parylene, or polyimide. Insulating layer can have a thickness ranging from 50 microns to 100 microns in one example.

[0049] Implantation of any of the prosthetic heart valves discussed herein inside of a patient (e.g., patient 2 shown in FIG. 1) can include the following steps. First, a sterilized prosthetic heart valve is crimped, from an assembled expanded state, using an appropriate crimping tool so that the prosthetic heart valve can be inserted into the delivery vehicle (e.g., expandable catheter). The crimped prosthetic heart valve can be inserted into the delivery site (e.g., mitral valve **18** shown in FIG. 2), and once properly positioned, can be re-expanded and sutured to surrounding tissue. In one example, the prosthetic heart valve can be oriented such that one sensor circuit is oriented along the coronal plane to face outward from the chest. Alternatively, a sensing circuit could be oriented along the sagittal plane such that it faces the left axilla of the patient. The dimensions of the pre-crimping (i.e., assembled) expanded state and the final (i.e., re-expanded) expanded state of the prosthetic heart valve can be substantially similar in one example. In an alternative example, the final expanded state of the prosthetic heart valve can be different (e.g., smaller) than the pre-crimping expanded state. Further, the deformable nature of the inductor coils of the various examples allows for crimping and re-expansion of the prosthetic heart valve with little to no change in self-resonance and no observed diminution of circuit performance.

[0050] Any of the various systems, devices, apparatuses, etc. in this disclosure can be sterilized (e.g., with heat, radiation, ethylene oxide, hydrogen peroxide, etc.) to ensure they are safe for use with patients, and the methods herein can comprise sterilization of the associated system, device, apparatus, etc. (e.g., with heat, radiation, ethylene oxide, hydrogen peroxide, etc.).

[0051] The treatment techniques, methods, steps, etc. described or suggested herein or in references incorporated herein can be performed on a living animal or on a non-living simulation, such as on a cadaver, cadaver heart, anthropomorphic ghost, simulator (e.g., with the body parts, tissue, etc. being simulated), etc.

Discussion of Possible Examples

[0052] The following are non-exclusive descriptions of possible examples of the present invention.

[0053] A prosthetic valve includes a flexible frame disposed along a frame axis and deformable about the frame axis between a crimped state and an expanded state. The frame includes a first end, a second end oppositely disposed from the first end, and a network of interconnected struts defining a plurality of cells. The prosthetic valve further includes a first circuit mounted on the frame. The first circuit includes a first inductor coil attached to and tracing a first subset of the struts such that the first inductor coil outlines a first subset of the plurality of cells, and a first sensor in electrical communication with the first inductor coil. The first sensor is configured to sense a physical parameter and generate a signal representing the physical parameter.

[0054] The prosthetic valve of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations, and/or additional components:

[0055] The prosthetic valve further includes a second circuit mounted on the frame. The second circuit includes a second inductor coil attached to and tracing a second subset of the struts such that the second inductor coil outlines a second subset of the plurality of cells, and a second sensor in electrical communication with the second inductor coil. The second sensor is configured to sense

the physical parameter and generate a signal representing the physical parameter.

[0056] The prosthetic valve further includes at least a first post assembly extending axially away from the first end, and at least a second post assembly extending axially away from the second end. The first sensor is mounted on the first post assembly, and the second sensor is mounted on the second post assembly.

[0057] The first post assembly comprises a first post and a first islet, the second post assembly comprises a second post and a second islet, the first sensor is mounted to the first islet, and the second sensor is mounted to the second islet.

[0058] The prosthetic valve further includes a first detuning mitigation layer disposed between the first inductor coil and the frame, and a second detuning mitigation layer disposed between the second inductor coil and the frame.

[0059] Each of the first and second detuning mitigation layers comprise ferrite.

[0060] The prosthetic valve further includes a first insulating layer surrounding the first detuning mitigation layer and the first inductor coil, and a second insulating layer surrounding the second detuning mitigation layer and the second inductor coil.

[0061] The first and second insulating layers comprise one of silicone, parylene, and polyimide.

[0062] Each of the plurality of cells has a pointed tip.

[0063] The first inductor coil is attached to the first subset of struts at the pointed tips of each of the first subset of the plurality of cells.

[0064] The first inductor coil is removably attached to the pointed tips of each of the first subset of the plurality of cells by a plurality of sutures.

[0065] Each of the plurality of sutures are formed from a biocompatible polymer material.

[0066] The second inductor coil is attached to the second subset of struts at the pointed tips of each of the second subset of the plurality of cells.

[0067] The second inductor coil is removably attached to the pointed tips of each of the second subset of the plurality of cells by a plurality of sutures.

[0068] Each of the plurality of sutures are formed from a biocompatible polymer material.

[0069] The first subset of the plurality of cells outlined by the first inductor coil comprises two cells in an axial direction, and three cells in a radial dimension.

[0070] The second subset of the plurality of cells outlined by the second inductor coil comprises two cells in the axial direction, and three cells in the radial dimension.

[0071] The frame is formed from a biocompatible metallic material.

[0072] Each of the first and second inductor coils are formed from gold.

[0073] Each of the first and second sensors are capacitive pressure sensors, and wherein the sensed physical parameter is pressure.

[0074] The first circuit has first self-resonant frequency ranging from 5 MHz to 50 MHz.

[0075] The second circuit has a second self-resonant frequency ranging from 5 MHz to 50 MHz, the second self-resonant frequency being different from the first self-resonant frequency.

[0076] The first circuit has first self-resonant frequency ranging from 10 MHz to 20 MHz.

[0077] The second circuit has a second self-resonant frequency ranging from 10 MHz to 20 MHz, the second self-resonant frequency being different from the first self-resonant frequency.

[0078] In the crimped state, the frame has a first axial dimension and a first radial dimension, and wherein in the expanded state, the frame has a second axial dimension and a second radial dimension.

[0079] The first axial dimension is greater than the second axial dimension, and the first radial dimension is smaller than the second radial dimension.

[0080] The frame is at least partially covered with a first biocompatible fabric, the first biocompatible fabric being disposed between at least a portion of the first inductor coil or the second inductor coil and the frame.

[0081] A second biocompatible fabric covers at least one of the first inductor coil or the second

inductor coil.

[0082] The prosthetic valve is implantable in a mitral valve of a patient.

[0083] In an implanted state of the prosthetic valve, the frame axis aligns with a flow of blood through the prosthetic valve.

[0084] The prosthetic valve is deliverable to the mitral valve of the patient via an expandable catheter.

[0085] The prosthetic valve is sterilized.

[0086] A method of implanting the prosthetic valve includes transitioning the prosthetic valve from the expanded state to the crimped state, delivering, via an expandable catheter, the prosthetic valve into the organ while in the crimped state, and returning the prosthetic valve to the expanded state once inside the organ of the patient.

[0087] The method of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations, and/or additional components:

[0088] In the crimped state, the frame has a first axial dimension and a first radial dimension.

[0089] In the expanded state, the frame has a second axial dimension and a second radial dimension.

[0090] The first axial dimension is greater than the second axial dimension, and the first radial dimension is smaller than the second radial dimension.

[0091] The organ is a heart.

[0092] The method further includes sterilizing the prosthetic valve prior to delivering the prosthetic valve to the organ.

[0093] A prosthetic valve assembly includes a prosthetic valve that includes a flexible frame disposed along a frame axis and deformable about the frame axis between a crimped state and an expanded state. The frame includes a first end, a second end oppositely disposed from the first end, and a network of interconnected struts defining a plurality of cells. The prosthetic valve further includes a first circuit mounted on the frame. The first circuit includes a first inductor coil attached to and tracing a first subset of the struts such that the first inductor coil outlines a first subset of the plurality of cells, and a first sensor in electrical communication with the first inductor coil. The first sensor is configured to sense a physical parameter and generate a signal representing the physical parameter. The prosthetic valve assembly further includes a transmitter in communication with the first sensor.

[0094] The prosthetic valve assembly of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations, and/or additional components:

[0095] The prosthetic valve further includes a second circuit mounted on the frame. The second circuit includes a second inductor coil attached to and tracing a second subset of the struts such that the second inductor coil outlines a second subset of the plurality of cells, and a second sensor in electrical communication with the second inductor coil. The second sensor is configured to sense the physical parameter and generate a signal representing the physical parameter.

[0096] The prosthetic valve further includes at least a first post assembly extending axially away from the first end, and at least a second post assembly extending axially away from the second end. The first sensor is mounted on the first post assembly, and the second sensor is mounted on the second post assembly.

[0097] The transmitter is in communication with the second sensor.

[0098] The prosthetic valve assembly further includes a power source in wired or wireless communication with the prosthetic valve assembly.

[0099] A monitoring system includes the prosthetic valve assembly and an external device in communication with the prosthetic valve assembly.

[0100] The monitoring system of the preceding paragraph can optionally include, additionally

and/or alternatively, any one or more of the following features, configurations, and/or additional components:

[0101] The external device comprises a transceiver in wireless communication with the prosthetic heart valve assembly.

[0102] The monitoring system further includes a remote monitor in communication with the prosthetic heart valve assembly via the external device.

[0103] A prosthetic valve includes a flexible frame disposed along a frame axis and deformable about the frame axis between a crimped state and an expanded state. The frame includes a first end, a second end oppositely disposed from the first end, and a network of interconnected struts defining a plurality of cells. The prosthetic valve further includes a multilayer sensing assembly mounted on the frame. The multilayer sensing assembly includes a first inductor coil pair comprising first upper and lower inductor coil portions, the first inductor coil pair disposed on a flexible substrate, a detuning mitigation layer disposed between the frame and the flexible substrate, and a first sensor in electrical communication with the first inductor coil pair. The first sensor is configured to sense a physical parameter and generate a signal representing the physical parameter.

[0104] The prosthetic valve of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations, and/or additional components:

[0105] The multilayer sensing assembly further includes a second inductor coil pair comprising second upper and lower inductor coil portions, the second inductor coil pair disposed on the flexible substrate, and a second sensor in electrical communication with the second inductor coil pair. The second sensor is configured to sense the physical parameter and generate a signal representing the physical parameter.

[0106] The detuning mitigation layer comprises ferrite.

[0107] The prosthetic valve further includes an insulating layer encapsulating the detuning mitigation layer and the flexible substrate.

[0108] The insulating layer comprises one of silicone, parylene, and polyimide.

[0109] A thickness of the insulating layer ranges from 50 microns to 100 microns.

[0110] The flexible substrate comprises polyimide.

[0111] A thickness of the flexible substrate ranges from 2 millimeters to 3 millimeters.

[0112] A thickness of the detuning mitigation layer ranges from 0.25 millimeters to 0.35 millimeters.

[0113] The prosthetic valve further includes a first soft magnetic layer disposed on the flexible substrate between the first upper and lower inductor coil portions, and a second soft magnetic layer disposed on the flexible substrate between the second upper and lower inductor coil portions.

[0114] The frame is formed from a biocompatible metallic material.

[0115] Each of the first and second inductor coil pairs are formed from one of gold, copper, and titanium.

[0116] Each of the first and second sensors are capacitive pressure sensors, and wherein the sensed physical parameter is pressure.

[0117] The prosthetic valve is implantable in a mitral valve of a patient.

[0118] The prosthetic valve is deliverable to the mitral valve of the patient via an expandable catheter.

[0119] The prosthetic valve is sterilized.

[0120] While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is

intended that the invention not be limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. A prosthetic valve comprising: a flexible frame disposed along a frame axis and deformable about the frame axis between a crimped state and an expanded state, the frame comprising: a first end; a second end oppositely disposed from the first end; and a network of interconnected struts defining a plurality of cells; a first circuit mounted on the frame, the first circuit comprising: a first inductor coil attached to and tracing a first subset of the struts such that the first inductor coil outlines a first subset of the plurality of cells; and a first sensor in electrical communication with the first inductor coil; wherein the first sensor is configured to sense a physical parameter and generate a signal representing the physical parameter.
2. The prosthetic valve of claim 1 and further comprising: a second circuit mounted on the frame, the second circuit comprising: a second inductor coil attached to and tracing a second subset of the struts such that the second inductor coil outlines a second subset of the plurality of cells; and a second sensor in electrical communication with the second inductor coil; wherein the second sensor is configured to sense the physical parameter and generate a signal representing the physical parameter.
3. The prosthetic valve of claim 2 and further comprising: at least a first post assembly extending axially away from the first end; and at least a second post assembly extending axially away from the second end; wherein the first sensor is mounted on the first post assembly; and wherein the second sensor is mounted on the second post assembly.
4. The prosthetic valve of claim 3, wherein: the first post assembly comprises a first post and a first islet; the second post assembly comprises a second post and a second islet; the first sensor is mounted to the first islet; and the second sensor is mounted to the second islet.
5. The prosthetic valve of claim 2 and further comprising: a first detuning mitigation layer disposed between the first inductor coil and the frame; and a second detuning mitigation layer disposed between the second inductor coil and the frame.
6. The prosthetic heart valve of claim 5, wherein each of the first and second detuning mitigation layers comprise ferrite.
7. The prosthetic heart valve of claim 6 and further comprising: a first insulating layer surrounding the first detuning mitigation layer and the first inductor coil; and a second insulating layer surrounding the second detuning mitigation layer and the second inductor coil.
8. The prosthetic heart valve of claim 7, wherein the first and second insulating layers comprise one of silicone, parylene, and polyimide.
9. The prosthetic valve of claim 2, wherein: the first inductor coil is attached to the first subset of struts at pointed tips of each of the first subset of the plurality of cells; the first inductor coil is removably attached to the pointed tips of each of the first subset of the plurality of cells by a first plurality of sutures; the second inductor coil is attached to the second subset of struts at the pointed tips of each of the second subset of the plurality of cells; and the second inductor coil is removably attached to the pointed tips of each of the second subset of the plurality of cells by a second plurality of sutures.
10. The prosthetic valve of claim 2, wherein the frame is formed from a biocompatible metallic material.
11. The prosthetic valve of claim 2, wherein each of the first and second inductor coils are formed from gold.
12. The prosthetic valve of claim 2, wherein each of the first and second sensors are capacitive pressure sensors, and wherein the sensed physical parameter is pressure.
13. The prosthetic valve of claim 2, wherein the first circuit has first self-resonant frequency

ranging from 5 MHz to 50 MHz, and wherein the second circuit has a second self-resonant frequency ranging from 5 MHz to 50 MHz, the second self-resonant frequency being different from the first self-resonant frequency.

14. The prosthetic valve of claim 2, wherein the frame is at least partially covered with a first biocompatible fabric, the first biocompatible fabric being disposed between at least a portion of the first inductor coil or the second inductor coil and the frame.

15. The prosthetic valve of claim 14, wherein a second biocompatible fabric covers at least one of the first inductor coil or the second inductor coil.

16. The prosthetic valve of claim 2, wherein the prosthetic valve is sterilized.

17. A monitoring system comprising: a prosthetic valve assembly comprising: the prosthetic valve of claim 2; a transmitter in communication with the first sensor and the second sensor; an external device in communication with the prosthetic valve assembly; wherein the external device comprises a transceiver in wireless communication with the prosthetic heart valve assembly.

18. A prosthetic valve comprising: a flexible frame disposed along a frame axis and deformable about the frame axis between a crimped state and an expanded state, the frame comprising: a first end; a second end oppositely disposed from the first end; a network of interconnected struts defining a plurality of cells; a multilayer sensing assembly mounted on the frame, the multilayer sensing assembly comprising: a first inductor coil pair comprising first upper and lower inductor coil portions, the first inductor coil pair disposed on a flexible substrate; a detuning mitigation layer disposed between the frame and the flexible substrate; and a first sensor in electrical communication with the first inductor coil pair; wherein the first sensor is configured to sense a physical parameter and generate a signal representing the physical parameter.

19. The prosthetic valve of claim 18, and further comprising: an insulating layer encapsulating the detuning mitigation layer and the flexible substrate.

20. The prosthetic valve of claim 18, and further comprising: a first soft magnetic layer disposed on the flexible substrate between the first upper and lower inductor coil portions.
